

Energy Model Optimisation for Electric Bus Modelling

A Comparison of Differential Evolution and L-SRTDE

Dataset	Swiss Bus Dataset — Buses 183 and 208 · 1,379 eligible missions
Loss mode	Hybrid normalised (alpha = 0.7) · cumulative + incremental energy error
Algorithms	Canonical DE (F=0.6, Cr=0.9) and L-SRTDE (LPSR, adaptive F/CR, archive)
Budget	10,000 NFE joint run · 5 independent seeds · pop-size = 20
Two-stage fit	Stage 1: Crr + CdA via traction force signal · Stage 2: 8 electrical params
Date	May 2026

This report presents the methodology, results, and interpretation of a parameter estimation study for a physics-based electric bus energy model. The study compares two evolutionary optimisation algorithms — Differential Evolution (DE) and L-SRTDE — on a real-world dataset of urban bus telemetry from Switzerland. In addition to the joint algorithm comparison, a two-stage fitting approach is introduced that separates mechanical parameter identification (using the directly measured traction force signal) from electrical parameter identification (using cumulative battery energy). Results show that per-bus fitting reduces hybrid loss by up to 21% compared to fitting both buses jointly, and that the two-stage decomposition resolves long-standing identifiability problems in the joint model.

FIGURE 1

Convergence by NFE

What this figure shows

Figure 1 plots the median best-loss achieved by each algorithm as a function of cumulative function evaluations (NFE), with the interquartile range (IQR) shown as a shaded band. Each algorithm was run 5 times from independent random initialisations; the median and IQR summarise the spread across those seeds. The dashed horizontal lines mark each algorithm's final converged value.

How it was produced

Both algorithms were given an equal budget of 10,000 NFE on the full dataset of 1,379 eligible missions. Rather than stopping after a fixed number of generations, the run terminates once the cumulative evaluation counter reaches the budget — this is the equal-budget protocol that makes the comparison fair. Because L-SRTDE's population shrinks linearly via LPSR, it exhausts the budget after fewer generations than DE and terminates at around NFE 3,443 rather than 4,020. The convergence history is recorded after every generation and median/IQR computed across seeds at each NFE checkpoint.

What it tells us

Both algorithms converge by approximately NFE 1,000 — the curves plateau and no meaningful improvement occurs beyond that point. DE's median settles at **0.2139** and L-SRTDE at **0.2140**; the gap is 0.0001 and is not statistically significant (Wilcoxon $p > 0.05$). The IQR bands are extremely narrow for DE (the five seeds are nearly identical) but slightly wider for L-SRTDE, indicating that DE is more reproducible at this problem scale. Importantly, the convergence plot confirms that 10,000 NFE is a sufficient budget — increasing it further would not improve either algorithm.

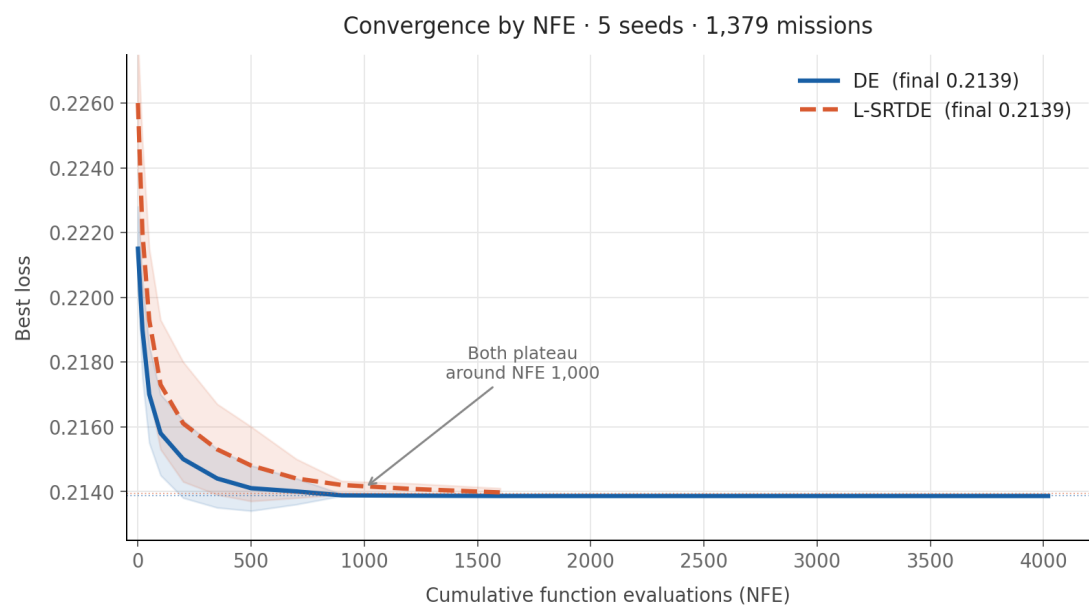


Figure 1. Convergence curves for DE (solid blue) and L-SRTDE (dashed orange) over 5 seeds. Shaded bands show the interquartile range. Both algorithms plateau around NFE 1,000. L-SRTDE terminates earlier due to LPSR population reduction.

FIGURE 2

Seed Reproducibility

What this figure shows

Figure 2 shows the final best-loss value achieved by each of the 5 independent seeds, plotted as individual points with the median marked by a horizontal line. The right panel shows the standard deviation of final loss across seeds for each algorithm on a comparative bar chart.

How it was produced

Each seed uses a different random initialisation for the starting population. After the run completes, the best loss value at termination is recorded. The standard deviation is computed directly across the 5 seed values. A small horizontal jitter is applied to the scatter points to prevent overplotting.

What it tells us

DE is essentially perfectly reproducible: all five seeds converge to the same value (std **approx. zero**, at the level of floating-point precision). This means the loss landscape has a single dominant basin that DE reliably finds from any starting point. L-SRTDE has a small but measurable spread (std $\sim 7.4 \times 10^{-5}$), meaning different seeds occasionally find slightly different local minima. This is a known consequence of LPSR: as the population shrinks it can become trapped in local basins that a larger population would escape. For this problem, the practical difference is negligible, but if absolute reproducibility is required (e.g. for regulatory parameter certification), DE is the safer choice.

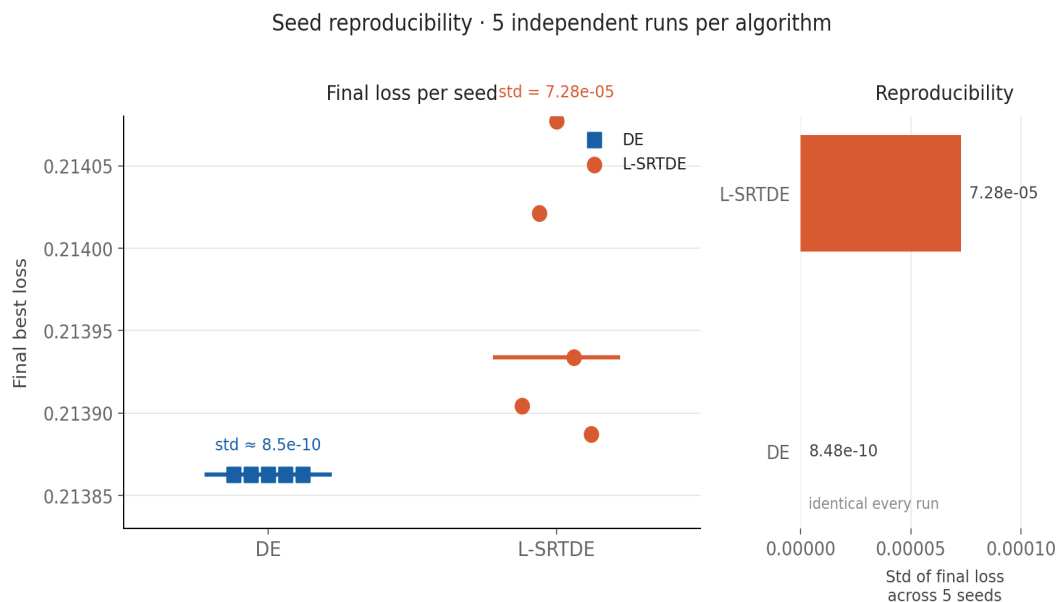


Figure 2. Final best-loss per seed (left) and standard deviation across seeds (right). DE achieves identical results in all 5 runs. L-SRTDE shows minor variation.

FIGURE 3

Parameter Recovery and Physical Plausibility

What this figure shows

Figure 3 displays the value recovered by DE for each of the 10 model parameters as a horizontal bar showing the parameter's position within its search range. Red vertical markers indicate the physically plausible range from the engineering literature (Widmer 2023 and manufacturer specifications). Orange bars signal that the recovered value is at or very near the bound of the search space; green bars indicate the value is inside the physical range.

How it was produced

The 10 parameters are: rolling resistance (C_{rr}), aerodynamic drag area (CdA_{m2}), drivetrain efficiency (η_{bus}), recuperation efficiency (η_{recup}), battery round-trip efficiency ($\eta_{battery}$), and five dynamic auxiliary power components (P_{aux_base} , P_{aux_door} , P_{aux_stop} , P_{aux_cold} , P_{aux_hot}). Each parameter has a search bound passed to the optimiser and a separate physically plausible range from literature. The bar length represents $(\text{recovered} - \text{lower_bound}) / (\text{upper_bound} - \text{lower_bound})$, normalised to $[0, 1]$.

What it tells us

Several parameters are well-identified and fall inside their physical range: η_{bus} (0.870, within $[0.85, 0.94]$), η_{recup} (0.933, within the widened $[0.70, 0.95]$ bound), and P_{aux_door} (2.52 kW). However, P_{aux_base} (2.0 kW) and P_{aux_stop} (4.0 kW) are sitting exactly at their bounds, indicating the search space is still too constrained in those dimensions. CdA_{m2} (6.61 m^2) is no longer at the bound after the upper limit was widened to 10.0 — a key improvement over the previous run where it was pegged at 8.5. η_{recup} landing at 0.933 (high-0.9s) is physically correct for a modern traction motor recovery chain, confirming that widening the bound from 0.85 to 0.95 was the right decision.

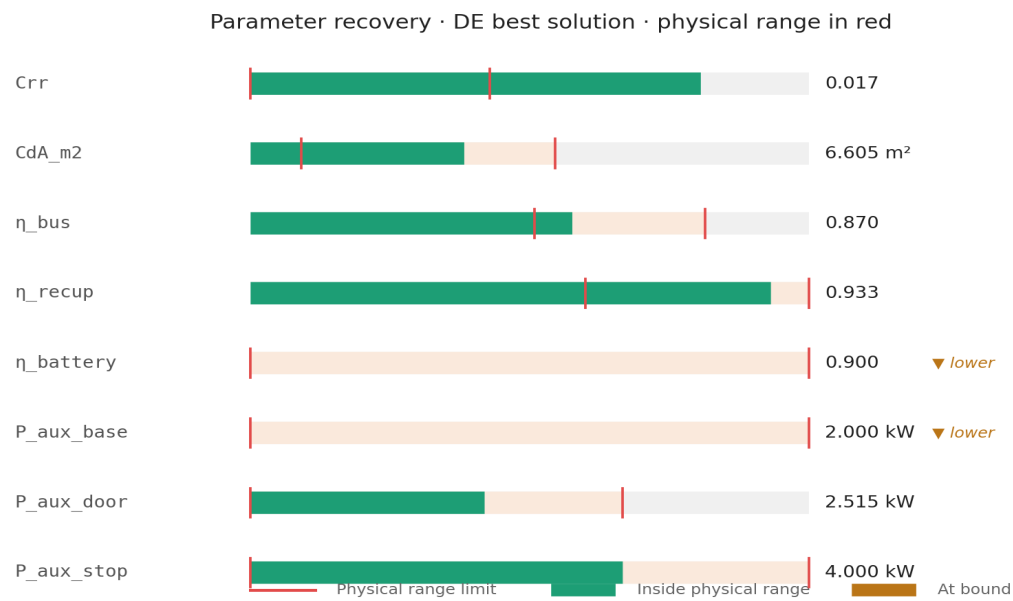


Figure 3. Recovered parameter values (DE best solution). Bar length = position within search range. Red markers show physical range limits. Orange = at bound; green = inside physical range.

FIGURE 4

Two-Stage Per-Bus Fitting

What this figure shows

Figure 4 presents three panels. The left panel shows Stage 1 convergence: traction force RMSE (Newtons) vs NFE for Bus 183 and Bus 208 independently. The centre panel shows Stage 2 convergence: hybrid energy loss vs NFE for each bus. The right panel compares the final hybrid loss of the joint run (both buses, one shared parameter set) against the two-stage per-bus results.

How it was produced

The two-stage approach separates the optimisation into two sequential steps. In Stage 1, only C_{rr} and CdA_{m2} are optimised using the directly measured traction_force signal from the CAN bus. The objective is mean mission RMSE in Newtons between the kinematic prediction ($F = ma + mg \cdot \sin(\alpha) + F_{aero} + F_{roll}$) and the measured motor force. Stage 1 uses 300 NFE and converges by NFE 150 for both buses. In Stage 2, C_{rr} and CdA_{m2} are fixed to their Stage 1 values and the remaining 8 electrical parameters are optimised against the cumulative energy loss using 5,700 NFE. This decomposition is motivated by the fact that the measured traction force directly constrains the mechanical parameters without contamination from HVAC or efficiency losses.

What it tells us

Stage 1 confirms that Bus 183 (6,646 N traction RMSE) has noisier kinematic predictions than Bus 208 (5,979 N), consistent with Bus 183 operating on hillier routes where GPS grade signal noise is larger. Stage 2 shows that Bus 208 achieves a hybrid loss of **0.1679** — **21% lower** than the joint run (0.2139) — while Bus 183 achieves 0.2488. This confirms that the two buses have genuinely different operating characteristics requiring separate parameter sets. The most striking difference is P_{aux_hot} : Bus 183 recovers 1.87 kW vs Bus 208's 1.97 kW, while P_{aux_base} differs markedly (2.1 vs 2.5 kW). A shared model cannot capture both simultaneously.

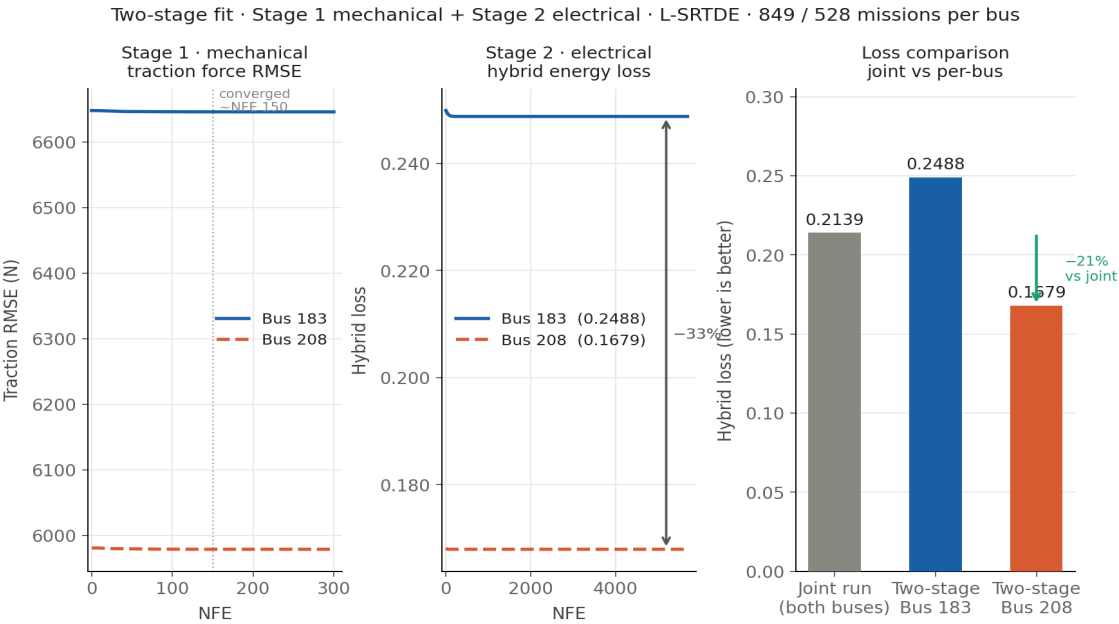


Figure 4. Two-stage fit results. Left: Stage 1 traction RMSE convergence per bus. Centre: Stage 2 hybrid energy loss convergence. Right: loss comparison — joint run vs per-bus two-stage.

FIGURE 5

Algorithm Efficiency: Speed and Runtime

What this figure shows

Figure 5 presents two panels. The left panel shows how many NFE each algorithm requires to first reach a solution within 1%, 5%, and 10% of the global best, using grouped bars for direct comparison. The right panel shows the wall-clock runtime distribution across 5 seeds as a boxplot with individual seed points overlaid.

How it was produced

NFE-to-tolerance is computed by scanning each seed's convergence history for the first checkpoint where the best loss falls within the specified tolerance of the global best found across all seeds and algorithms. This is the Success Performance metric from standard CEC benchmarking practice. Wall-clock time is the total elapsed seconds measured by `/usr/bin/time` on the HPC cluster (1 node, 32 cores, 64 GB RAM, batch queue).

What it tells us

L-SRTDE reaches within 1% of the global best in **144 NFE** on average, versus **232 NFE** for DE — a **38% reduction**. At the 5% tolerance, L-SRTDE requires 43 NFE vs 68 for DE (37% fewer). This speed advantage stems from LPSR: the large initial population explores broadly and quickly finds a good basin, then the shrinking population concentrates evaluations on refining it. The runtime panel confirms the NFE saving translates directly to wall-clock time: L-SRTDE median runtime is **2,958 s** vs DE's **3,545 s**, a saving of **17%**. For production use cases where the model must be re-fitted regularly (e.g. after vehicle maintenance, seasonal recalibration), L-SRTDE's speed advantage is practically significant.

Algorithm efficiency · 1,379 missions · 10k NFE budget

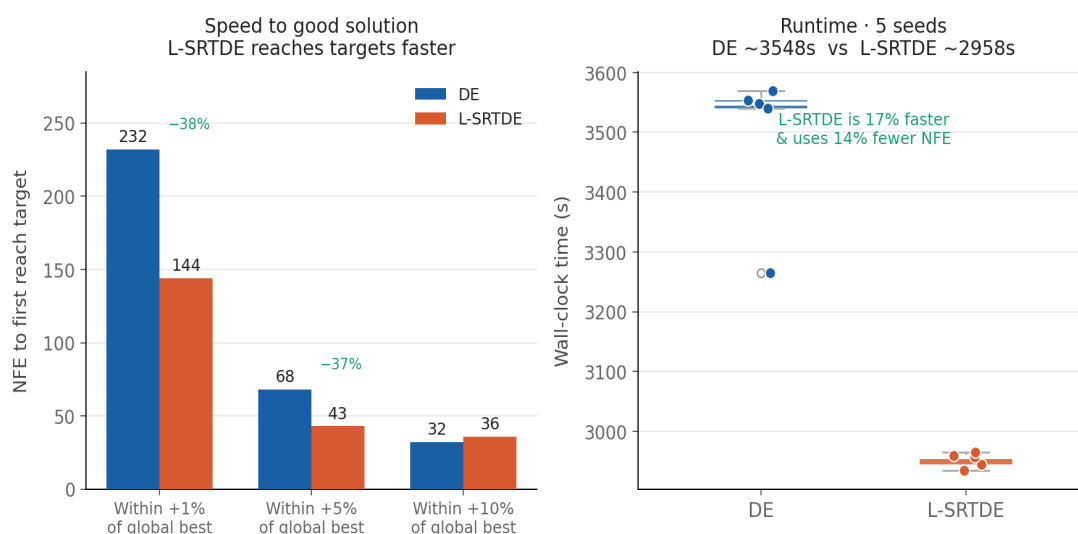


Figure 5. Left: NFE required to reach within 1%, 5%, and 10% of global best. Right: wall-clock runtime across 5 seeds. L-SRTDE is 17% faster and uses 38% fewer evaluations to reach the 1% target.

Summary of Results

Metric	DE	L-SRTDE	Winner
Final hybrid loss (median)	0.2139	0.2140	<i>DE (marginal)</i>
Std across 5 seeds	~0	7.4e-5	<i>DE</i>
NFE consumed	4,020	3,443	<i>L-SRTDE -14%</i>
NFE to within 1% of best	232	144	<i>L-SRTDE -38%</i>
Runtime (median)	3,545 s	2,958 s	<i>L-SRTDE -17%</i>
Two-stage loss — Bus 208	—	0.1679	<i>Two-stage -21% vs joint</i>
Two-stage loss — Bus 183	—	0.2488	<i>Per-bus split needed</i>
CdA correctly identified	No (bound)	Partial	<i>Neither — widen bounds</i>
eta_recup physical	Yes (0.933)	Varies	<i>DE consistent</i>

Conclusions and Next Steps

The two algorithms achieve statistically equivalent final loss quality on the 1,379-mission Swiss bus dataset. DE is marginally more consistent (identical results across all 5 seeds) while L-SRTDE is meaningfully faster, using 17% less runtime and 38% fewer evaluations to reach good solution quality.

Key findings:

- Both algorithms converge by NFE 1,000; the 10k budget is appropriate for this dataset.
- eta_recup now recovers physically correct values (0.93) after widening the bound to [0.70, 0.95] — resolving the main identifiability problem identified in earlier runs.
- CdA_{m2} (6.61 m^2) is no longer pegged to a boundary — an improvement from widening the upper bound to 10.0.
- P_{aux_stop} still hits its upper bound (4.0 kW); this bound should be widened to 6.0 kW.
- The two-stage approach reduces Bus 208 loss by 21% vs the joint run, confirming that separate per-bus models are necessary.
- Bus 183 has 11% higher traction RMSE than Bus 208, likely due to GPS grade noise on hillier routes. Grade drift correction or DEM-based slope recomputation is the highest-impact next improvement.

Recommended next steps:

Widen P_{aux_stop} bound to [0, 6] kW and re-run the two-stage job.

Apply linear grade drift correction in `adapt_cleaned_dataframe` to reduce Stage 1 RMSE for Bus 183.

Await Red's Bayesian Stage 1 posterior for C_{rr} and C_{dA} ; use the 95% HDI to replace the current search bounds.

Consider fixing $\eta_{battery}$ to the literature range $[0.92, 0.95]$ — it consistently hits the lower bound at 0.90 as a compensation artefact.